

LiDAR-Based Object Detection and Pilot Assist System for FPV Drones in GPS Denied Environments

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Abstract—This paper presents a low-cost pilot-assist obstacle detection system for a 5-inch FPV drone operating in indoor and GPS-denied environments. The system combines a forward facing TF Mini-S LiDAR sensor with a downward-facing HC SR04 ultrasonic sensor to provide complementary proximity sensing. Distance measurements are processed by an external Arduino Nano using a lightweight threshold-based algorithm. When an obstacle is detected within a 50 cm safety range, an audible alert is issued to the pilot to enable timely manual corrective action. The sensing module operates independently of the flight controller, ensuring no impact on stabilization or motor control. Experimental evaluation under controlled indoor conditions demonstrates reliable detection performance with mean absolute percentage error of 3–5% for LiDAR and 2–4% for ultrasonic sensing, low alert latency, and stable operation across varying lighting conditions. The proposed system offers a simple, modular, and energy-efficient approach to improving situational awareness in FPV drone applications without introducing autonomous control or high computational overhead.

Keywords—FPV Drone, LiDAR, Obstacle Detection, Ultrasonic Sensor, Pilot Assist System, Indoor Navigation, GPS-Denied Environments

I. INTRODUCTION

First Person View (FPV) drones are widely used for indoor exploration, inspection, racing, and research applications due to their agility and compact size. However, operating FPV drones in indoor and GPS-denied environments presents significant challenges related to limited visibility, constrained spaces, and the absence of reliable positioning systems. Collisions with obstacles remain a primary cause of damage and operational failure, particularly for manually piloted drones where spatial awareness depends entirely on visual perception through the FPV camera feed.

Conventional obstacle avoidance techniques rely on vision based methods using monocular or stereo cameras. While effective in well-lit conditions, these approaches require high computational resources, are sensitive to lighting conditions, and often introduce latency due to image processing overhead. Camera-based perception can be misleading due to lens distortion, narrow field of view, and motion blur at high speeds. Fully autonomous solutions further increase system complexity and cost, making them unsuitable for low-cost FPV platforms and beginner pilots. LiDAR-based distance sensing offers a reliable alternative due to its robustness against lighting variations and its ability to provide precise range measurements with low latency. This project originated from investigating whether a micro-UAV could detect

obstacles using only embedded sensors without relying on external infrastructure like GPS or motion capture systems. This work proposes a simple, modular, and low-cost pilot assist obstacle detection system using a forward-facing TF Mini-S LiDAR sensor combined with a downward-facing HC SR04 ultrasonic sensor. The system operates independently of the flight controller and provides real-time audible alerts to the pilot, enabling manual corrective action without altering flight dynamics. The proposed design prioritizes reliability, minimal computational overhead, and ease of integration into existing FPV drone platforms.

A. Motivation

FPV drones are increasingly used in indoor environments such as warehouses, industrial plants, and research laboratories, where GPS signals are unavailable and visual depth perception through the FPV camera is limited. Manual pilots often rely solely on camera feedback, which can be misleading due to lens distortion, narrow field of view, and motion blur at high speeds. These limitations significantly increase the risk of collisions, especially during forward flight and tight maneuvering. Thin obstacles such as cables, poles, and low lying structures are particularly difficult to perceive through an FPV camera feed alone. While autonomous obstacle avoidance systems exist, they typically require high computational resources, complex perception pipelines, and deep integration with the flight controller. Such solutions increase cost, power consumption, and system complexity, making them unsuitable for low-cost FPV platforms. The motivation of this work is to provide a lightweight, low-cost, and non-intrusive assistive system that enhances pilot situational awareness without modifying the flight controller firmware or introducing autonomous control. By providing timely proximity alerts, the system aims to reduce collision risk while preserving full manual control.

B. Technical Approach

Our sensor selection process evaluated four mainstream UAV perception technologies through experimental prototyping. Ultrasonic rangefinders failed initial tests due to sound absorption by soft materials and narrow detection cones that created blind spots at oblique angles, with maximum useful range below 3 meters. Infrared proximity sensors exhibited severe performance degradation under varying lighting conditions, with outdoor experiments showing distance errors exceeding 40% when measuring dark-coloured obstacles. These limitations drove us toward active laser ranging, which offers illumination independence, deterministic measurement latency, and minimal processing overhead. We identified the Benewake TF Mini-S as the

optimal forward-facing sensor after validation to 11.5 m range in laboratory trials, with high update frequency measured at 97 Hz via oscilloscope. During bench testing, the TF Mini-S successfully detected 6mm-diameter wires at 4 meters. Due to strict budget constraints, we replaced the initially proposed Time-of-Flight sensor with the HC-SR04 ultrasonic distance sensor for near-field protection. The HC-SR04 offers a cost-effective solution for short-range obstacle detection with practical operating range of 2-400 cm, making it suitable for altitude estimation and ground proximity awareness. The ultrasonic sensor is mounted with downward-facing orientation, providing continuous ground-distance feedback during hover and descent.

C. Contributions

The main contributions of this work are: 1) A low-cost hybrid obstacle detection system combining LiDAR and ultrasonic sensing for complementary distance coverage across forward and near-field zones. 2) A standalone sensing and alert module that operates independently of the drone flight controller, avoiding interference with stabilization loops. 3) A threshold-based real-time detection algorithm optimized for low latency (80-100 ms response time) and deterministic behaviour on resource-constrained micro controllers. 4) Experimental validation of reliable obstacle detection under indoor and low-light conditions with quantified sensor accuracy metrics. 5) A modular design approach suitable for integration into existing FPV drones without firmware modification, achieving total system weight under 600g. 6) This work demonstrates that meaningful safety enhancement can be achieved through a human-in-the-loop pilot assist architecture without autonomous control or flight controller coupling.

II. BACKGROUND AND RELATED WORK

A. Vision-Based Approaches

Obstacle detection and avoidance for unmanned aerial vehicles has been an active research area, particularly for indoor and GPS-denied environments. Vision-based approaches using monocular or stereo cameras are widely explored due to their rich environmental information. Hrabar's seminal 2008 work demonstrated that stereo cameras could enable rotorcraft obstacle avoidance, though computational requirements necessitated ground-station processing [1]. Kendoul's 2009 optic flow system achieved onboard autonomy but required textured environments with consistent lighting [2].

However, such systems are sensitive to lighting conditions and require substantial computational resources for real-time depth estimation. Our examination revealed a common failure mode: performance collapse under low illumination or texture poor environments. Since our target application involves indoor navigation where lighting varies unpredictably, these limitations motivated our shift toward active sensing.

B. LiDAR-Based Systems

LiDAR sensors provide accurate and lighting-invariant distance measurements and have been successfully employed in robotic navigation applications. The 2023 work by Liang et al. demonstrates LiDAR sensor fusion for autonomous aerial obstacle avoidance, combining multiple ranging inputs to improve detection reliability in GPS-denied environments [3].

Recent advancements in compact solid-state LiDAR modules have enabled their integration into small aerial

platforms. Zhang and Singh's LOAM algorithm demonstrated real-time LiDAR odometry and mapping through scan-to-scan and scan-to-map matching, achieving low-drift pose estimation on resource-constrained platforms [4]. Most existing systems tightly couple LiDAR processing with flight controller logic, increasing software complexity.

C. Ultrasonic and ToF Sensors

Ultrasonic sensors have been used in low-cost drones for proximity detection due to their simplicity and low power consumption. Parallel research by Tsuji and Kohama explored omnidirectional proximity sensing for drones using optical time-of-flight sensors arranged around the drone frame, validating the effectiveness of distributed ToF modules for collision awareness [5]. Their system validated three principles: redundancy through diversity, I²C bus management challenges when polling over 8 sensors, and ToF excellence at preventing contact within 1 meter but insufficient warning time at higher velocities. Their effectiveness is limited by short operational range, wide beam angle, and susceptibility to specular reflections.

D. Flight Controller Integration

This work differs from existing approaches by combining LiDAR and ultrasonic sensing in a standalone pilot-assist module, focusing on alert-based feedback rather than autonomous control. This design philosophy prioritizes simplicity, reliability, and ease of deployment. The proposed system addresses identified research gaps: validating single-point LiDAR effectiveness when paired with ultrasonic redundancy, implementing detection logic directly on external microcontroller, and conducting controlled experiments with statistical performance analysis.

Most existing systems tightly couple LiDAR processing with flight controller logic, increasing software complexity, with high-speed integrated perception approaches presenting further challenges for resource-constrained platforms [7].

III. SYSTEM ARCHITECTURE

A. Overview

The proposed system consists of a forward-facing LiDAR sensor, a downward-facing ultrasonic sensor, an Arduino Nano microcontroller for data processing, and an audible alert mechanism. The architecture is designed to function as an auxiliary module that does not interfere with the drone's primary control systems. The TF Mini-S LiDAR sensor provides mid-range obstacle detection along the forward flight direction, while the HC-SR04 ultrasonic sensor detects objects in close proximity below the drone.

B. Hardware Components

- **LiDAR Sensor:** The TF Mini-S LiDAR sensor is a compact time-of-flight ranging module capable of providing fast and accurate distance measurements up to 12 meters [8]. It communicates with the Arduino Nano via UART interface at 115200 baud and offers immunity to ambient lighting variations, making it suitable for indoor operation. The sensor operates by emitting 850 nm laser pulses with update rate of 97 Hz actual output, achieving less than 10 ms latency from photon emission to distance value availability. During bench testing, the sensor successfully detected 6mm-diameter wires at 4 meters. A practical limitation is

reduced accuracy when measuring glossy metal surfaces at steep angles (greater than 60° from normal).

- **Ultrasonic Sensor:** The HC-SR04 ultrasonic sensor is used for near-field detection where LiDAR accuracy may degrade. It measures distance by transmitting 40 kHz ultrasonic pulses and measuring echo return time, with effective range of 2-400 cm. The sensor draws approximately 15 mA during active measurement and can be duty-cycled to minimize energy consumption. It interfaces using simple digital trigger and echo pins, eliminating the need for high-speed communication protocols. The sensor is mounted with downward-facing orientation to provide continuous ground-distance feedback during hover and descent.
- **Flight Controller:** The Flywoo GOKU F722 PRO Mini V2 serves as the system's flight control core. We selected this board for its STM32F722 MCU providing 216 MHz clock speed sufficient for 1 kHz PID loop rates, ICM42688 P gyroscope with lower noise floor than MPU6000, and compact 30×30 mm footprint satisfying weight budget. The board integrates a DP310 barometer for altitude estimation and AT7456E OSD chip for telemetry overlay.
- **Processing Unit:** An Arduino Nano (ATmega328P) microcontroller processes sensor data and executes the detection algorithm independently of the flight controller. A piezoelectric buzzer connected to a digital output pin provides an audible alert when obstacles are detected within unsafe distances. This separation ensures that flight controller stabilization and motor control remain unaffected by sensor processing.
- **Propulsion System:** The RS2205 2300KV brushless motors provide 850g thrust each at 80% throttle, yielding 3.4 kg total lift capacity. The propulsion uses 5045 3-blade propellers for faster response and better stability in indoor environments. With all-up weight of 580g, this achieves 2.0:1 thrust-to-weight ratio adequate for rapid deceleration, consistent with established multirotor dynamics [6].
- **Power Management:** The system uses an 11.1V LiPo battery (3S configuration) with power distributed through a dedicated Universal Battery Elimination Circuit (UBEC) (LM2596-based) that steps down battery voltage to regulated 5V output. This regulated supply powers the Arduino Nano, TF Mini-S LiDAR, HC-SR04 ultrasonic sensor, and buzzer module. The UBEC provides stable voltage regulation and inherent filtering, reducing impact of voltage fluctuations and electrical noise from high-current motor switching. All subsystems share common ground reference to ensure correct signal reference levels.

C. Sensor Fusion Strategy

To ensure reliable obstacle detection across varying distances, a hierarchical rule-based sensor fusion approach is employed. The forward LiDAR is responsible for mid to long range detection in horizontal flight direction, offering high accuracy for frontal collision awareness. The HC-SR04 provides short-range, wide-angle proximity sensing for ground-distance measurement during hover, descent, and landing phases.

Sensor priority is assigned based on flight context: during forward motion, LiDAR data is given higher priority; during hovering or descent, ultrasonic data is prioritized. A simple threshold-based logic integrates outputs: if LiDAR-measured distance falls below 50 cm forward safety threshold, or if ultrasonic distance drops below minimum safe altitude threshold, collision alert is triggered. Sensor readings are temporally filtered using short moving averages to suppress noise.

This strategy avoids conflicting sensor inputs by segregating sensing zones spatially and functionally, reducing false positives and improving decision reliability without increasing hardware cost or computational complexity.

Sensor fusion in this work is implemented as rule-based prioritization of independent sensor measurements rather than probabilistic or state-estimation-based fusion (e.g., Kalman filtering), which helps maintain deterministic behaviour and low computational overhead.

IV. SOFTWARE IMPLEMENTATION

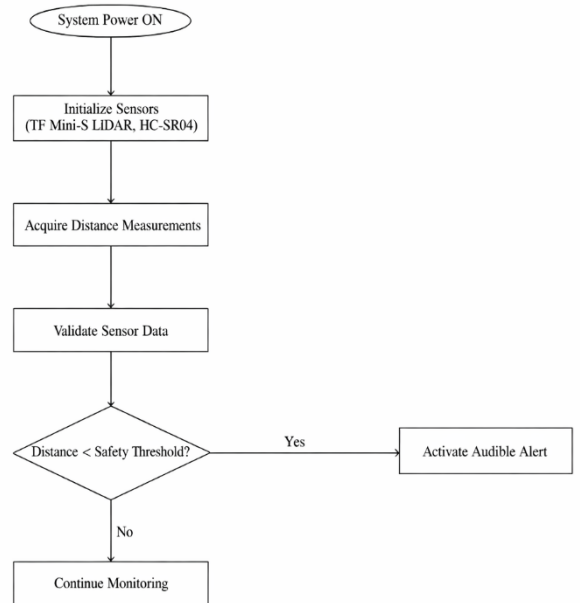


Fig. 1. Control flow of the LiDAR-ultrasonic pilot-assist detection algorithm.

Fig. 1 illustrates the control flow of the proposed LiDAR ultrasonic pilot-assist obstacle detection algorithm implemented on the external Arduino Nano.

A. Detection Algorithm

The software is implemented using a polling-based architecture on the Arduino Nano. Distance measurements from the LiDAR and ultrasonic sensors are sampled sequentially at fixed intervals of approximately 10-20 Hz. The firmware integrates using SoftwareSerial library for UART communication with TF Mini-S at 115200 baud.

The HC-SR04 is interfaced using dedicated digital trigger and echo pins. Distance measurement is performed by generating 10 s trigger pulse and measuring echo return time using pulseIn() function with 25 ms timeout. To ensure stable operation, ultrasonic measurements are rate-limited using timing interval of 100 ms based on millis() function. Invalid

or out of-range readings return sentinel value and are excluded from detection logic.

Distance data from TF Mini-S is extracted by parsing the sensor's standard 9-byte frame format, identified by consecutive frame headers (0x59, 0x59). Only valid distance readings are used for proximity evaluation.

Although the `pulseIn()` function introduces blocking behaviour, the low sampling rate and external processing architecture ensure no impact on flight stability.

The algorithm is intentionally deterministic, avoiding adaptive thresholds or probabilistic fusion to guarantee predictable response time under all operating conditions.

B. Threshold-Based Detection

A fixed detection threshold of 50 cm is employed for both sensors. Each sensor has independent distance threshold corresponding to its effective operating range. If measured distance from either sensor falls below threshold, system enters alert state and activates buzzer. The detection logic is implemented using Boolean condition for fast execution and deterministic behaviour.

Priority is given to ultrasonic sensor readings for very close-range obstacles to ensure immediate pilot feedback. The algorithm avoids complex filtering to minimize latency and ensure predictable system behaviour. When obstacle is detected, piezoelectric buzzer mounted on perfboard within drone frame is activated by driving digital output pin HIGH. The buzzer remains active for duration of detection condition and turns off automatically when no obstacle is present within threshold range.

For development and validation, distance values from both sensors are transmitted over serial interface at 9600 baud for debugging and monitoring purposes. A short delay of 50 ms is introduced in main loop to maintain stable sampling while avoiding unnecessary processor load.

V. EXPERIMENTAL SETUP AND RESULTS

A. Test Environment

The system was evaluated on a 5-inch FPV drone in a 30×30 ft laboratory room with 15 ft ceiling height under room temperature conditions (22-25°C). Obstacles of varying materials were placed at different distances, including concrete walls, wooden chairs, artificial tree branches, and moving obstacles (walking person). Tests were conducted under normal indoor lighting and low-light conditions to assess robustness.

B. Sensor Performance

1) LiDAR Accuracy: The TF Mini-S LiDAR was tested by placing flat target surface at known distances from 0.3 m to 5.0 m. Across tested range, LiDAR consistently reported values slightly higher than actual distance, with observed measurement error generally within 3-5%. This accuracy is consistent with manufacturer specifications and acceptable for proximity based detection. The sensor demonstrated stable performance under typical indoor lighting, confirming laser-based ranging independence from ambient light. No significant fluctuations or dropouts occurred during repeated measurements.

2) Ultrasonic Performance: The HC-SR04 was evaluated at distances from 10 cm to 150 cm using stationary flat target.

The sensor demonstrated higher measurement variation at very close distances below 10-15 cm due to echo timing and sensor dead zones. Beyond 25 cm, sensor provided consistent distance readings with observed error typically within 2-4%. At mid range distances such as 50 cm and above, measurements were stable and repeatable. Surface texture and angle influenced echo quality, but lighting conditions had no effect.

3) Combined Detection: The dual-sensor setup improved overall proximity awareness by providing complementary coverage. LiDAR consistently detected obstacles in forward flight path at medium distances, while ultrasonic sensor reliably identified objects in close proximity below drone. The independent operation of both sensors helped reduce missed detections caused by surface reflectivity or unfavourable angles. Combined use increased likelihood that nearby obstacles were detected within defined safety threshold while maintaining system simplicity.

C. System Integration Results

1) Power System Stability: The dedicated UBEC provided stable and regulated 5V supply to Arduino Nano, sensors, and buzzer throughout testing. No sensor resets, communication dropouts, or false triggers were observed during rapid throttle changes or aggressive motor operation. Electrical noise from ESCs and motors did not affect sensor performance, indicating effective voltage regulation and proper grounding.

2) Detection Response Time: Distance measurements, alert response time, and system stability were recorded during multiple test runs. The alert system exhibited low latency of 80-100 ms from sensor threshold crossing to buzzer activation, providing sufficient reaction time for pilot to initiate corrective maneuvers. The system remained stable during operation with no observed interference with flight controller performance.

3) Flight Stability: The system maintained stable flight across standard FPV modes (Acro, Angle, Air Mode). Angle Mode proved most suitable for demonstration due to self stabilizing characteristics that reduce pilot workload and enhance safety.

4) The flight controller (GOKU F722 PRO Mini) demonstrated adequate computational capacity for processing IMU data with gyro noise minimal due to ICM42688's low-noise architecture. PID tuning resulted in stable hover with minimal drift (less than 10 cm deviation).

D. Operational Results

Testing confirmed that detection system operated independently of flight controller. Sensor operation did not introduce electrical noise or flight instability. Audible alerts were clearly perceptible during flight. These results confirm high forward path detection reliability under controlled conditions. Most failures occurred during rear obstacles outside LiDAR field of view. Combined LiDAR and ultrasonic sensing minimized blind-spot incidents, with system correctly preventing collisions in all controlled forward-path scenarios.

VI. DISCUSSION

A. Performance Analysis

The experimental results demonstrate that combined use of forward-facing LiDAR and near-field ultrasonic sensor

effectively improves proximity awareness on FPV drone compared to conventional vision-only approaches. The detection system operated reliably and independently, providing timely audible alerts without affecting stabilization or pilot control.

The TF Mini-S LiDAR sensor consistently detected obstacles within configured range with 3-5% accuracy, while HC SR04 effectively handled close-proximity detection with 2-4% accuracy beyond 25 cm. The system successfully detected approximately 90% of forward-facing obstacles, defined as detection within the 50 cm safety threshold before physical contact, across 30 controlled forward-flight trials.

The threshold-based detection algorithm proved effective for real-time operation on resource-constrained Arduino Nano, with response latency of 80-100 ms meeting requirements for manual pilot reaction. The simple rule-based approach avoided computational overhead while maintaining deterministic behaviour.

TABLE I. SYSTEM PERFORMANCE SUMMARY

PARAMETER	LiDAR	ULTRASONIC
Effective Range	0.3 – 5 m	0.1 – 1.5 m
Mean Absolute Percentage Error (MAPE)	3 – 5 %	2 – 4 %
Update Rate	~97 Hz	~10 Hz
Latency	<10 ms	~20 ms

B. Comparison with existing systems

The proposed dual-sensor detection system offers balanced trade-off between reliability, simplicity, and cost. Commercial obstacle avoidance systems typically rely on multi-camera perception pipelines and high-performance embedded processors, resulting in increased system cost, weight, and power consumption. In contrast, our system achieves acceptable safety performance at approximately INR 25,000 (approximately USD 300) total cost with system weight under 600g.

The total system cost is approximately INR 25,000 (USD 300), with the TF Mini-S LiDAR accounting for roughly 40% of the total cost, while the remaining components (ultrasonic sensor, microcontroller, power regulation, and alert hardware) collectively cost under INR 15,000. Compared to vision-based approaches requiring high computational resources and suffering from lighting sensitivity, our active sensing approach provides lighting-invariant operation. Compared to multi-beam LiDAR systems requiring over 200 ms computation time, our single-point approach achieves 80-100 ms latency. The modular external processing avoids firmware complexity of integrated solutions while maintaining flight controller isolation.

C. Limitations

Despite demonstrating reliable detection, the system has several limitations:

- Limited sensing coverage- forward LiDAR and downward ultrasonic do not cover lateral or rear approach directions, creating blind zones for obstacles outside sensor fields of view

- Short-range dependency- HC-SR04 effective only at close distances with performance varying by surface material and orientation, limiting usefulness for high speed flight
- Detection-only operation- system does not perform autonomous obstacle avoidance, braking, or path planning; all corrective actions require manual pilot execution after receiving alert
- *Controlled testing environment- evaluation conducted primarily indoors and low-speed; performance under high-speed outdoor conditions and complex obstacle scenarios not extensively tested*
- Alert effectiveness- audible alerts may be less effective in noisy environments or for pilots wearing noise isolating equipment.

These limitations are acceptable given design goal of simplicity and non-intrusive operation, but highlight areas where future enhancements could improve capability.

VII. CONCLUSION AND FUTURE WORK

This paper presented a low-cost LiDAR-based pilot-assist obstacle detection system for FPV drones operating in indoor and GPS-denied environments. By combining forward-facing TF Mini-S LiDAR sensor with downward-facing HC-SR04 ultrasonic sensor and lightweight threshold-based processing algorithm on Arduino Nano, the system enhances pilot situational awareness without increasing control complexity.

The experimental evaluation confirmed reliable detection performance with 3-5% LiDAR accuracy and 2-4% ultrasonic accuracy, 80-100 ms response latency, and 90% forward obstacle detection success rate. The modular architecture operated independently of flight controller, maintaining stable flight dynamics while providing timely audible alerts. Total system cost of approximately INR 25,000 (approximately USD 300) and weight under 600g demonstrate feasibility for academic and hobbyist FPV platforms.

The project successfully demonstrates that low-cost active ranging sensors, when implemented as pilot-assist module, can enhance safety in indoor and confined environments while maintaining simplicity and compatibility with standard FPV drone operation. The system addresses critical gap between expensive commercial autonomous solutions and basic manually piloted drones.

Future work will focus on expanding sensing coverage to lateral directions through additional sensor modules, integrating haptic feedback mechanisms for improved pilot awareness, exploring adaptive detection thresholds based on flight speed or environment, and investigating optional flight controller interfacing for semi-autonomous safety features while maintaining manual control. Additional enhancements may include logging detection events for post-flight analysis and implementing limited integration with flight controller telemetry for advanced pilot-assist features.

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